

Goji Dynamic Liquidity, Concept Design (v2)

Document: 01 Concept Design · **Version:** 2.0 (complete rewrite, supersedes v0.1 and the original whitepaper framing) **Status:** For internal review **Owner:** Goji Crypto Holdings LLC
Feeds into: 02 Detailed Design, 03 Technical Specification, 04 API Specification

1. What this is, in one paragraph

Goji Dynamic Liquidity is a dynamically managed, onchain synthetic index. The index token (HANU) is paired against a set of constituent assets, each pairing forming its own liquidity pool. HANU's market price is the arbitrage merged, depth weighted blend of those pairings. An AI policy continuously rotates the blend between risk constituents (for example ETH, BTC, LINK) and a defensive stablecoin sleeve (USDC), based on trend signals across several timeframes. The objective is to manage the volatility and reduce the drawdown of the index relative to a static hold of the same constituents. It is not a way to fix, stabilize, or guarantee HANU's price.

2. The mechanism (validated model)

2.1 Price formation

- Each HANU/X pairing is a separate pool. Absent trades, the implied HANU price in a pool moves with its counter asset X.
- Arbitrage merges every pool's implied price into a single market price for HANU.
- Therefore HANU's price is roughly the depth weighted blend of whatever it is currently paired to. Shifting depth between constituents shifts the blend.

2.2 The rotation (the core operation)

To move exposure from a volatile constituent toward the defensive sleeve: 1. Withdraw an increment of liquidity from the HANU/X pool. Proportional withdrawal returns both legs (HANU and X) and does not move HANU's price. It reduces that pool's depth. 2. Sell only the withdrawn X for USDC. 3. Redeposit the withdrawn HANU and the new USDC into a HANU/USDC pool.

Net effect: the same HANU is now paired to a stablecoin instead of a volatile asset, which removes that constituent's beta from the blend. The reverse operation re acquires risk exposure.

2.3 Two hard invariants

- Never sell HANU. HANU is never sold into a pool. Rotations touch only the counter asset legs. The withdrawn HANU is redeposited, not sold.
- No direct HANU trades. The operator does not buy or sell HANU to move its price. HANU's price is set by external arbitrage. The operator influences it only by where it

places depth. This is a deliberately weaker and more defensible lever than direct market making. See Section 8.

2.4 What the system is, named honestly

This is tactical asset allocation, or trend following, with a cash sleeve. Its closest traditional analog is a managed futures or time series momentum strategy. It has two sources of return and risk: - Time series decision. In (risk) versus out (USDC). Drives drawdown reduction. - Cross sectional decision. Which constituents to hold when risk on. Drives relative performance.

3. Performance character (what to actually expect)

The two halves of the trade are independent and have different reliability:

Half	Drives	Reliability	Failure mode
Exit (de-risk into a downturn)	Drawdown reduction	Robust, the easy half	Late exit gives back the first leg down
Re-entry (re-risk into a recovery)	Recovery capture and outperformance	Hard, the whole edge lives here	Early or false re-entry re-acquires beta and erases the banked advantage

Because a successful exit preserves value through a crash, a well timed re-entry rides the recovery from a higher base than a static hold, so the upside is outperformance, not parity. But the bottom is a hindsight label and no signal targets it reliably. The honest summary is that the entire profit and loss lives at the turns, and the product is a bet on turn timing edge. The drawdown reduction benefit, which comes from the exit alone, is the robust floor under that bet.

Known failure regimes (must be modeled in 02): - Whipsaw and chop. Non trending markets generate false signals, and rotation costs bleed with no trend to capture. - Sharp V recoveries. Confirmation based re-entry lags the bounce, and in a fast V the lag is most costly. - Reserve exhaustion. The USDC sleeve defends HANU's price against arbitrage sell pressure out of finite reserves. A defense that outlasts reserves breaks. The stable phase is a reserve funded defense, not free. - Execution drag. Each rotation increment is several onchain transactions (withdraw, swap, add), exposed to gas spikes and MEV exactly when markets move fast.

4. Objectives and non-goals

Goals

- G1. Reduce maximum drawdown of the HANU index versus a static hold of its constituents.

- ## Non-goals (first class, every design decision is checked against these)

- ## 5. Volatility dial (the corrected mental model)

Stability and “we picked the asset that went up” are different products with different risk. The dial makes the tradeoff explicit and measurable, and it replaces the old, incorrect stable price framing.

6. Conceptual architecture

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+-----+
kill      |                                     |6 Human oversight,
switch    |                                     |
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1. Data ingestion. Market, onchain, leverage and short interest, sentiment (weighted low), macro. Live today in data collection mode.
2. Signal engine. Trend confirmation across timeframes (see Section 7), anomaly detection, per pool risk scoring.
3. Policy engine. Maps signals to a target allocation across constituents and USDC. Emits proposals only.
4. Risk and guardrails. Validates every proposal against hard constraints, enforces the two invariants, trims or rejects.
5. Execution. Performs rotations as tranches, protects against slippage and MEV, idempotent.
6. Human oversight. Approvals above thresholds, instant halt, and a kill switch that freezes execution while sensing continues.

7. Signals and policy

- Decision horizon. Primary signals on the 1 day and 3 day frames. The 3 hour and 9 hour frames are used as faster confirmation only. This cadence keeps rotations to a handful per month, which is what makes the execution cost model viable.
- In and out decision (time series). Use trend confirmation, meaning price versus a moving average, moving average slope, or the sign of multi period momentum. Do not drive in and out with oscillators such as RSI or Stochastic. Those are mean reversion signals and they structurally fight a trend system, manufacturing whipsaws.
- Re-entry rule. Re-enter on trend reconfirmation, accepting a lag, rather than on oversold. Oversold re-entry is a falling knife rule and is the expensive failure mode in Section 3.
- Constituent selection (cross sectional). Relative strength, or cross sectional momentum, among candidate assets when risk on.
- Oscillators (RSI and similar). Retained only for position sizing nuance, never for the regime flip.
- Confluence. Act when the daily and 3 day frames agree. This trades lag for far fewer false signals and is the main tuning dial.

RSI in earlier discussion was an illustrative placeholder. The real stack is the trend confirmation design above.

8. Risk, guardrails, and the legal shape

Guardrails (defaults to be finalized in 02)

- Max rotation per step (tranche sizing, for example 5, 10, or 20 percent of a pool).
- USDC reserve floor sufficient to fund the intended defense depth.
- Slippage caps and pre-trade simulation on every rotation.
- Per constituent exposure caps, and cooldowns between rotations.
- Kill switch and instant mode downgrade.

Legal shape (flag, not advice)

A token whose value is the manager's AI growing and protecting a basket NAV is, in substance, a managed fund or decentralized investment fund interest, not a utility token. The corpus's own Part 1 paper already uses that term. The two invariants (never sell and no direct trade) materially improve the posture by removing direct price manipulation, but they do not change the fund likeness. This must be resolved with securities counsel before any external publication or marketing. The non-goals exist to keep the build on the right side of that line.

9. Operating modes (evidence gated autonomy)

Mode	Sense	Decide	Execute	Human	State
M0 Data collection	yes	logged	no	builds baseline	today
M1 Advisory	yes	proposals	no	executes manually	next
M2 Guarded autonomy	yes	yes	yes, small, within guardrails	approves above limits	later
M3 Full autonomy	yes	yes	yes, within guardrails	exception only	aspirational

Each step up requires a met success metric over a defined window, a guardrail review, and operator sign off. Downgrade is instantaneous.

10. Open questions for the Detailed Design

- OQ1. Exact trend confirmation rules and the daily and 3 day confluence logic.
- OQ2. Tranche sizing and rotation cadence per volatility regime.
- OQ3. USDC reserve floor policy and the reserve exhaustion guardrail.

- OQ4. MEV and slippage protection strategy for rotation execution.
 - OQ5. Constituent set and cross sectional selection method.
 - OQ6. Roles, if any, of MIA, GOJ, and GABA in the index.
 - OQ7. Multi chain scope (Base, Solana, Arbitrum, Aptos, Optimism per the corpus) and per chain execution differences.
 - OQ8. Backtest methodology and the success metrics that gate each mode transition.
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Sign-off checklist

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Section 2 invariants (never sell HANU, no direct HANU trades) confirmed as hard rules.

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Objective confirmed as drawdown and volatility management, not price stability (NG1).

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Current mode confirmed as M0.

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Section 8 legal shape item routed to counsel.

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OQ1 through OQ8 assigned for the Detailed Design.